Zakaria Baannou

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Education

International University of Rabat (UIR) — Metropolitan Area, Rabat
Engineer's Degree In Computer Science (Big Data & AI Specialization) — *Sept. 2020 – August 2025*

Experience

IT/Software Consultant at XLA Corporation — September 2025 – Present
Enabling clients to best use their IT infrastructure and improving workflow efficiency. — *Rabat, Morocco*

Contributor at AtlasIA — September 2025 – Present
Building a diffusion LLM for the Moroccan dialect, and contributing to various projects. — *Remote; Paris, France*

Research Assistant — June 2025 – Present
New York University — *Remote; New York City, USA*

- Researching applications of LLMs and foundational time series models in FOREX markets under the supervision of Prof. Dennis Shasha
- Developing experiments and running them on HPC
- Writing down concise notes, in addition to storytelling-based presentations

AI Research Engineer Intern (Capstone Project) — February 2025 – August 2025
Omniops — *Rabat, Morocco*

- Independently prototyped a residual architecture on GPT-2 that improved convergence; the mechanism was concurrently validated by Meta FAIR ('Canon Layers') to enable **length generalization without positional encoding** and **double reasoning performance** at scale
- Engaged in technical correspondence with Dr. Zeyuan Allen-Zhu regarding parameter initialization details in the footnotes, receiving recognition for identifying theoretical nuances overlooked by broader readership
- Designed and ran ablations on GPT2 pretraining on Openwebtext, to test the horizontal residual links

Data Scientist Intern — July 2024 – August 2024
DXC Technology — *Rabat, Morocco*

- Preprocessed and vectorized logs using Log2Vec
- Applied an unsupervised learning approach with an AutoEncoder model to detect anomalies with an F1 score of 0.97 in a testing environment
- Deployed and managed services with Docker and FastAPI, integrated by configuring Zabbix Agent settings

Technician Assistant — June 2022 – July 2022
Ibn Sina University Hospital Center — *Rabat, Morocco*

- Assisted with moving servers while minimizing downtime and troubleshooting network issues brought by staff
- Performed assessment and troubleshooting of computer and network issues brought by staff
- Discovered the national network's architecture, and learned about the tools used to maintain and secure it

MoroccoAI Summer School For Multimodal Foundation and Gen. AI — September 2025
Out of 1300+ applicants, 60 students, professors and engineers were selected to attend. — *Rabat, Morocco*

1st Place Winner in the "48 Hours to Bring Ideas to Life" challenge — December 2023
As a team of 10 students, we pitched an innovative solution to improve a golfer's experience. — *Rabat, Morocco*

2nd Place Team, E-Tofoula Hackathon For Children's Mental Health — February 2024
Developed & pitched ITYHAD. Later presented to Princess Lalla Meryem and Mrs. Brigitte Macron. — *Rabat, Morocco*

Publication

ForExAI: Time Series Inference and News Article Analysis Reveal Profitable Foreign Exchange Signals. Mathematics, 14(13), 2319.

Projects

Ad Supervision Data Pipeline | *Spark, Kafka, Airflow, Presto, PostgreSQL, Redis, Metabase*

- Engineered a real-time data streaming pipeline using Kafka and Spark Streaming to capture mock API events and continuously ingest structured data into PostgreSQL.
- Orchestrated automated batch ETL workflows with Apache Airflow and Spark Batch to extract, process, and sync historical data across parallel Redis and PostgreSQL stores.
- Implemented a Lambda architecture by unifying real-time and batch streams via Presto for distributed SQL querying, powering interactive Metabase dashboards to monitor click-through, conversion rate and other KPIs.

Vision Transformer For Uni-Modal Speech Emotion Recognition | *HuggingFace, PyTorch, Librosa, W&B*

- Utilized librosa to extract a Mel-spectrogram from the Toronto emotional speech set (TESS)
- Compared Performance between a finetuned Whisper-small & ViT B16
- Obtained Whisper-small levels of performance at **35.49% the model size**, and **2.66% the inference time**

Public Sentiment Analysis on Generative AI | *Python, Jupyter, HuggingFace, Tensorflow, PraW, Google API*

- Collected data from Reddit and Youtube comments alongside university campus and amphitheaters
- Prepared the data for quantitative analysis, applied hypothesis testing to find possible correlations
- Displayed visualizations for the statistical findings and their interpretations using Dash in addition to pyLDAvis

Clothing Type Detector | *Python, Pillow, Tensorflow, Javascript, Azure, DockerHub, GitHub*

- Trained customvision.ai's model on images of 10 types of clothing, modified the code to trigger a Serverless Function when predicting a class, containerized the model
- Deployed on an Azure Container App, that stores classified images in Blob Storage using a Serverless Function
- Developed a web page to use the model through a vanilla JS API

Investigation of Hallucinations in LLama 70B using Model Steering | *NNsight, Repeng, Transformers, PyTorch*

- Investigated possibility of changing behaviors using neuronpedia's web GUI
- Generated a synthetic dataset of expected vs. steered behavior using Gemini 3 for the mech. interp. tools
- Currently exploring different methods of training control vectors for steering

Technical Skills

Languages: Python, Javascript, C, SQL, noSQL (MongoDB & Neo4j)
Frameworks: FastAPI, Springboot, Dash, Arduino
Big Data Tools: Apache Hadoop, Pig, Airflow, NiFi, Kafka, Spark, Hive, Presto, PowerBI, Metabase
Developer Tools: Git, Docker, Azure, Jupyter, Conda, UV
Libraries: Pandas, PyTorch, Keras, Accelerate, Transformers, ElasticSearch, NLTK, Plotly

Languages

Arabic : Proficient **French** : Proficient **English** : Fluent